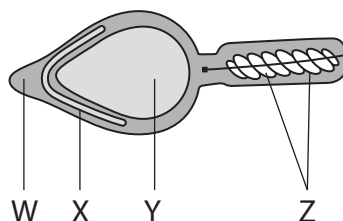


8 The phenotype of a rabbit is affected by its genotype.

Rabbit sperm cells contain genetic information that will become part of the genotype of the offspring.

The diagram shows part of a rabbit sperm cell.



- (a) Which labelled structure contains the genetic information used in the synthesis of acrosomal enzymes?

(1)

- ☐ A W
- ☐ B X
- ☐ C Y
- ☐ D Z

- (b) A fertilised rabbit egg cell will divide by mitosis to form a solid ball of genetically identical cells.

Name this solid ball of cells.

(1)



- (c) The genetically identical cells in a developing rabbit embryo will continue to divide by mitosis.

Some of the cells produced can undergo epigenetic modification as they develop into specialised cells.

- (i) Epigenetic modification can alter the activation of certain genes.

Describe **one** form of epigenetic modification that could occur in the rabbit cells.

(2)

- (ii) Explain how cells in the rabbit embryo become specialised after epigenetic modification.

(3)



*(d) The table shows information about three types of rabbit.

Type of rabbit	Black 	Himalayan 	Albino 
Colour of fur	Solid black all over	White fur with black ears, nose, paws and tail	White all over
Allele combination(s)	CC, Cc^h, Cc^a	$c^h c^h, c^h c^a$	$c^a c^a$

Enzymes produced by the rabbits control the production of pigments in the fur.

One type of rabbit produces an enzyme that is inactive at the core body temperature of the rabbit.



Discuss how genes and the environment interact to produce the fur colour of these three types of rabbit.

Use the information in the question to support your answer.

(6)

(Total for Question 8 = 13 marks)

TOTAL FOR PAPER = 80 MARKS

